

$$\text{Beta} = 1$$

$$\sigma_{\text{market}} = 0.20$$

$$\sigma_{\text{stock}} = 0.30$$

$$\sigma_{\varepsilon} = \sqrt{0.30^2 - 0.20^2}$$

R_m = simulated market returns following random walk with 5000 trades in the day with volatility scaled for one day and one trading period, i.e.,

$$\sigma_{\text{market}} \text{ Per Trade Period} = 0.20 \cdot \frac{1}{\sqrt{250}} \cdot \frac{1}{\sqrt{5000}}$$

The simulation process is as follows:

- Step 1.** Specify the customer order size and side. For example, in iteration 1 specify a buy order for 200,000 shares of RLK. This represents 20% ADV and will consist of approximately 1000 customer trades since stock RLK has an average trade size of 200 shares. Analysts performing this simulation exercise will change the order size and side in each iteration.
- Step 2.** Simulate 5000 trades from other market participants and the number of trades from the simulated order. Sequence the customer trades throughout the day following any preferred methodology. For example, sequence customer trades over the day following a VWAP strategy or an aggressive front-loaded strategy. Customer order trades should be alternated with market participant trades in a random fashion but following the specified strategy. We encourage analysts to experiment with various sequencing schemes to simulate different trading algorithms. For the first iteration there will be 6000 trades in total: 5000 from other market participants and 1000 from the customer order.
- Step 3.** Simulate market prices for these orders:

$$P_t = P_{t-1} + (MI(\text{Shares}(t)) \cdot \text{Side}(t)) - 0.95 \cdot (MI(\text{Shares}(t-1)) \cdot \text{Side}(t-1)) + \text{Beta} \cdot R_m + \xi_t$$

where $\xi_t \sim N(0, \sigma_{\varepsilon})$ and $t = 1$ to 6000. Notice that in the formulation above we add in the full market impact of the trade and subtract 95% of the impact from the previous trade. This accounts for the dissipation of temporary impact—we assume an immediate dissipation of 95%. The 95% factor is consistent with our findings in Chapter 5. Readers are encouraged to experiment with various different temporary percentages and market impact